

PRE-PURCHASE INSPECTION

Expedition Vehicle Inspection Checklist

Use this checklist before buying any expedition truck, camper, or overland vehicle. Take it with you. Fill it in on site. Do not rely on memory.

7 sections · 72 inspection points · Verdict framework included

From people who bought the wrong vehicle once, then the right one.

VEHICLE DETAILS

Complete this section before beginning the inspection. Keep a copy for your records.

Vehicle Make / Model:	Year / Chassis No.:
Asking Price:	Location:
Inspection Date:	Inspector Name:

HOW TO USE THIS CHECKLIST

▣ Tick each item when inspected and satisfactory. ▣ Red-flag hints show what failure looks like. ▣ Write observations in the notes field — exact details matter when negotiating.

Bring a torch, magnet, and OBD reader. Hire a specialist mechanic for any serious candidate. Never inspect a vehicle you're emotionally attached to — take someone rational.

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PAPERWORK & HISTORY

Get these before you even look at the vehicle. If a seller can't produce them, that's your first red flag.

☐ Full service history and maintenance records

▣ Gaps > 1 year are a concern. Look for continuity — same workshop stamps are a good sign.

☐ VIN / chassis number matches all documents

▣ Mismatched numbers = potential fraud, stolen vehicle, or cut-and-shut. Walk away immediately.

☐ Owner history — how many previous owners?

▣ More than 3–4 in 5 years suggests a problem vehicle that keeps being passed on.

☐ No accident, flood, or fire damage history

▣ Ask directly. In Europe, check with national databases (HPI equivalent in your country).

☐ Registration documents, weight certificate, Certificate of Conformity

▣ Missing CoC or weight cert can make the vehicle impossible to register or insure correctly.

☐ TÜV / roadworthy certificate validity (if in Europe)

▣ Freshly renewed TÜV doesn't mean all is well — sellers sometimes do minimal work to pass.

☐ Receipts for major work claimed (engine, gearbox, tyres)

▣ Verbal claims of "just rebuilt engine" mean nothing without paperwork. Ask for invoices.

☐ Outstanding finance check (HPI or equivalent)

▣ If the vehicle has outstanding finance, the lender can repossess it even after you buy it.

☐ Odometer reading matches service stamps and wear patterns

▣ Worn steering wheel + fresh 40,000 km reading = clocked. Check seat bolster wear too.

Inspector notes:

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CHASSIS & FRAME

Get under the vehicle with a torch and magnet. This inspection cannot be done from standing height.

- ☐ **No cracks or heavy rust on ladder frame rails**
▮ Surface rust is normal. Flaking, pitting, or cracks near welds are serious — expensive to repair properly.
- ☐ **Crossmembers in good condition — no cracks or repairs**
▮ Look for fresh weld beads that don't match the original metal. Someone patched something.
- ☐ **All suspension mounts and bushings sound**
▮ Torn or collapsed bushings = vague handling, uneven tyre wear, and noise on rough tracks.
- ☐ **Axle condition — no cracks, no oil leaks from diff or hubs**
▮ Axle oil leaks will contaminate brakes. A leaking diff left unaddressed damages internals.
- ☐ **Underbody — brake lines, fuel lines, and wiring harness intact**
▮ Chafed wiring is a fire risk. Corroded or kinked brake lines are a safety failure.
- ☐ **Body mounts and sub-frame (or iso-mount system) in good condition**
▮ Cracked or loose body mounts transfer road stress into the shelter body — causes delamination over time.
- ☐ **Recovery points, tow hitch, and underbody skids secure**
▮ Loose recovery points are dangerous. Bent skid plates indicate previous hard contact.
- ☐ **Jacking points accessible and undamaged**
▮ In a field breakdown, you need to be able to safely lift the vehicle. Verify before you need them.

Inspector notes:

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ENGINE & DRIVETRAIN

Inspect cold first, then after a full warm-up. Some problems only appear at operating temperature — or only when cold.

- ☐ **Cold start — no excessive smoke, no rough running**
▮ Brief white puff on cold diesel start = normal. Sustained blue (oil) or white (coolant) smoke = serious.
- ☐ **Oil colour and level — check for contamination**
▮ Milky/creamy oil = coolant mixing in (head gasket or cracked block). This is expensive. Walk away or price in.
- ☐ **Coolant condition — check colour and for oil contamination**
▮ Oily coolant = same problem from the other side. Dark/sludgy coolant = long deferred maintenance.
- ☐ **Compression or leak-down test (strongly recommended)**
▮ The single best indicator of engine health. Ask seller permission; any refusal is a red flag.
- ☐ **Turbo — no shaft play, no oil leaks, no whine or surge**
▮ Check intake hose for oil mist (blown turbo seal). Turbochargers are expensive to replace.
- ☐ **Injectors, belts, hoses — condition and last service date**
▮ Timing belt/chain failures are catastrophic. If it's overdue, it needs to be priced into the deal.
- ☐ **Gearbox and transfer case shift smoothly through all gears**
▮ Crunching, grinding, or hard shifts = worn synchros or selector forks. Get a quote before buying.

☐ **4WD system and all diff locks engage and disengage cleanly**

▮ *Test on a loose surface, not just tarmac. Diff locks that bind or stick are expensive to repair.*

☐ **Prop shafts — no excessive play in universal joints**

▮ *Worn UJs cause vibration and can shear at high speed. Check by rocking the prop shaft by hand.*

☐ **No unusual noises, rattles, or vibrations at any speed**

▮ *Note the exact condition when it occurs — cold/warm, loaded/unloaded, speed range.*

Inspector notes:

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BODY & SHELTER

Especially important for FM2, Dornier, SGH, and other aluminium shelter bodies. Water ingress ruins insulation and is hard to trace.

☐ **Aluminium sandwich panels — no delamination, bubbling, or soft spots**

▮ *Press panels firmly. Delamination feels spongy and is a significant repair. FM2 is generally excellent here.*

☐ **All corner extrusions and rivets intact — no missing or pulled rivets**

▮ *Missing rivets indicate previous repair or flex cracking. Check external corners carefully.*

☐ **Roof integrity — no dents, cracks, or pooling water**

▮ *Even small roof cracks let water in. Pooled water stains on the ceiling indicate a leak.*

☐ **Floor condition — no soft spots, rot, or moisture damage**

▮ *In aluminium bodies, floor damage is usually impact-related. In steel bodies, check for rot.*

☐ **Door and window seals — all seal flush with no gaps**

▮ *Old or cracked seals are cheap to replace but indicate overall maintenance standard.*

☐ **Water leak test — hose test or check after rain**

▮ *Run a hose over the roof and all seams while someone inspects inside. Catches what visual inspection misses.*

☐ **External storage lockers — seals, hinges, locks all functional**

▮ *Stiff or rusted locks become inaccessible in the field. Test every single one.*

☐ **Ventilation and heating system (diesel heater, roof vent) functional**

▮ *A faulty Webasto/Espar in January is not a comfort issue — it's a safety issue.*

☐ **Insulation — no moisture damage or compression**

▮ *Damaged insulation means you'll freeze in winter and bake in summer. Difficult to retrofit.*

Inspector notes:

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ELECTRICAL SYSTEM

Amateur wiring is the most common hidden problem on expedition builds. Take your time here.

☐ **Main and leisure battery condition — load tested (not just visual)**

▮ *A battery can read 12.6V and still fail under load. Use a proper battery tester or ask the seller.*

- ☐ **Alternator output and charging system functioning correctly**
▮ Rev the engine and check voltage at the leisure battery — should read 13.8–14.4V when charging.
- ☐ **All lights, switches, gauges, and instruments working**
▮ Non-functional gauges (especially oil pressure, temp, fuel) are safety-critical. Budget for repair.
- ☐ **Wiring quality — no exposed splices, no cable ties holding wire bundles together**
▮ This is where fires start. Look for melted insulation, overloaded fuses, and abandoned wiring runs.
- ☐ **Fuse box — no blown fuses, no bypassed fuses, no signs of heat**
▮ Bypassed fuses (wire where a fuse should be) are extremely dangerous. This is a hard no.
- ☐ **Solar panels (if fitted) — clean, uncracked, and controller functioning**
▮ Check controller display shows correct charge state. Cracked cells still look fine but perform poorly.
- ☐ **Inverter and shore power system (if fitted) functional**
▮ Test the inverter under load — run a kettle. Shore power inlet should be weatherproof and secure.
- ☐ **Ground / earth points clean, tight, and corrosion-free**
▮ Poor grounds cause mysterious electrical faults. Check the engine earth strap first.

Inspector notes:

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INTERIOR & LIVABILITY

You're going to live in this. Check it the way you'd check a rental apartment — every drawer, every tap, every corner.

- ☐ **Layout works for your crew size and mission**
▮ Stand in it. Lie in the bed. Open every storage compartment. Is it actually going to work day-to-day?
- ☐ **Ventilation — roof vent / fan operational and sealing correctly**
▮ A non-sealing roof vent lets rain in. A seized fan is a hot, unhappy camper in summer.
- ☐ **Heating system fully functional and recently serviced**
▮ Ask for a full hot-start test on the diesel heater. Cold-start is easy; hot-restart after shutdown is the real test.
- ☐ **Water pump working and all plumbing connections leak-free**
▮ Run the pump for 3 minutes and check every joint. Small leaks rot floors and breed mould over months.
- ☐ **Kitchen — cooktop, oven (if fitted), and sink drain all functional**
▮ Test every burner. A faulty gas regulator or blocked drain is a daily annoyance on the road.
- ☐ **Fridge / cooling system operational**
▮ Run the compressor for 20 minutes. If it's a 3-way fridge, test all three modes (12V, 240V, gas).
- ☐ **Fresh water and grey water tanks — no cracks, leaks, or odour**
▮ Smell matters. A tank that has held stagnant water for months may need professional cleaning.
- ☐ **Storage — volume adequate and weight distribution sensible**
▮ Heavy items (tools, water, spare parts) should be low and centred. Check what's currently stored where.
- ☐ **Interior surfaces — no mould, damp patches, or unexplained staining**
▮ Mould on the ceiling means condensation or a leak. Either needs to be resolved before the first trip.

Inspector notes:

Drive it hard — that's what you'll be doing. If the seller won't allow a proper test drive, that tells you everything.

- ☐ **Engine temperature reaches and holds normal operating range**
Temp climbing above normal or fluctuating under load = cooling system issue. Stop immediately.
- ☐ **Steering — no pull, no wander, appropriate effort**
Pull under braking = warped disc or uneven tyre pressure. Wander = worn steering components or geometry.
- ☐ **Brakes — firm pedal, no fade, straight stopping**
Brake fade after 3–4 stops = glazed pads or overheated fluid. Budget for a full brake service.
- ☐ **Clutch — smooth engagement throughout the range**
Slipping at high RPM under load, or judder from cold = clutch near end of life. Get a quote.
- ☐ **No significant vibration at any speed, especially motorway speed**
Vibration at specific speeds = out-of-balance tyres or bent prop shaft. High-speed test is essential.
- ☐ **4WD and all diff locks tested under actual load — not just engaged**
Lock them on a dirt track or grass, then accelerate, brake, and corner. Faults only show under load.
- ☐ **Articulation on uneven ground — no binding, no unusual movement**
Watch all four corners over a speed bump at an angle. Any corner that lifts unexpectedly = suspension issue.
- ☐ **Test fully loaded if possible — or ask specifically about behaviour when heavy**
A truck that drives fine empty can handle completely differently at its rated gross weight.
- ☐ **Parking brake holds on an incline**
Many expedition trucks have weak parking brakes on steep slopes. Test it. You will need it.

Inspector notes:

FINAL VERDICT

- ☐ **PASS — Ready to buy.**
All checks satisfactory. Proceed with confidence.
- ☐ **PASS WITH CONDITIONS**
Minor issues identified — negotiate price or agree repairs before completing purchase.
- ☐ **MAJOR REPAIRS NEEDED**
Significant issues found — buy only if price is deeply discounted and you fully understand the repair scope.
- ☐ **WALK AWAY**
Red flags too significant. Cut your losses and keep looking.

Overall observations / negotiation notes:

RECOMMENDED

Always hire a qualified expedition mechanic or specialist workshop for a full PPI on any serious candidate. The cost (typically €200–€500) is the best insurance you can buy. A mechanic who knows these vehicles will spot things this checklist cannot.

The best inspection is the one you actually do on site.

Don't trust memory. Don't trust the seller's description. Don't trust how good it looks in photos.

Print this checklist. Take it with you. Fill every box.

Safe travels & keep the shiny side up.

— Jason & Mareike · [3xploreWithUs](#)

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